WS: 10

Date: 12-2-2025 Lec # 13 Day: M T W T F S

Disk = Circle (x and y plane)

D = Determinant

Second Derivative Test:-

Suppose the second partial derivatives of f are continuous on a Disk with centre (a,b), and suppose that $f_x(a,b) = 0$ and $f_y(a,b) = 0$.

[So (a,b) is a critical point of f].

Let $D = D(a,b) = f_{xx}(a,b)\, f_{yy}(a,b) - [f_{xy}(a-b)]^2$

a) If $D > 0$ and $f_{xx}(a,b) > 0$, then $f(a,b)$ is local minimum.

b) If $D > 0$ and $f_{xx}(a,b) < 0$, then $f(a,b)$ is local maximum.

c) If $D < 0$; then (a,b) is a saddle point of f.

Note 1:-

If $D = 0$, the test gives no Informations f could have a local maximum or local minimum at (a,b) or (a,b) could be a saddle point of f.

Note 2:-

To remember the formula for D its helpful to write it as a determinant:

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Formula:-

$$D = \begin{vmatrix} f_{xx}(a,b) & f_{xy}(a,b) \\ f_{yx}(a,b) & f_{yy}(a,b) \end{vmatrix} = f_{xx}(a,b) f_{yy}(a,b) - (f_{xy}(a,b))^2$$

Exp 1:- Find the local maximum, ~~and~~ minimum ~~values~~, and saddle points of $f(x,y) = x^4 + y^4 - 4xy + 1$.

Solution:-

first we find saddle points / critical points

$f_x = 0$	and $f_y = 0$
$4x^3 + 4y = 0$	$= 4y^3 - 4x = 0$
$4(x^3 - y) = 0$	$= 4(y^3 - x) = 0$
$= x^3 - y = 0$	$= \boxed{y^3 - x = 0} \to$ eq ②
$\cancel{-}y = \cancel{-}x^3$	
$\boxed{y = x^3} \to$ eq ①	

Now put eq ① int eq ②

$= y^3 - x = 0$

$= (x^3)^3 - x = 0$

$= x^9 - x = 0$

$= x(x^8 - 1) = 0$

$= x((x^4)^2 - (1)^2) = 0$

$= x((x^2)^2 - (1))(x^4 + 1) = 0$

$= x(x^2 - 1)(x^2 + 1)(x^4 + 1) = 0$

Here

$x = 0$, $x^2 - 1 = 0$, $x^2 + 1 = 0$, $x^4 + 1 = 0$

$\boxed{x = 0}$

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$$x^2 - 1 = 0$$
$$x^2 = 1$$
$$x = \sqrt{1}$$
$$\boxed{x = \pm 1}$$

$$x^2 + 1 = 0$$
$$x^2 = -1$$
$$\boxed{\sqrt{x^2} = \sqrt{-1}} \longrightarrow \text{Complex number.}$$

$$x^4 + 1 = 0$$
$$x^4 = -1$$
$$\boxed{\sqrt{(x^2)^2} = \sqrt{-1}} \longrightarrow \text{Complex number}$$

Now, the real values of x are

$$\boxed{x = 0, 1, -1}$$

Put $x = 0, 1, -1$ in eq ①.

$x = 0$ in eq ①

$$y = x^3 = 0^3 = \boxed{0}$$

First Saddle point is $(0, 0)$

Now put $x = 1$ in eq ①

$$y = x^3 = (1)^3 = \boxed{1}$$

Second saddle point is $(1, 1)$

Now put $x = -1$ in eq ①

$$y = x^3 = (-1)^3 = \boxed{-1}$$

Third saddle point is $(-1, -1)$

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So, Saddle points are :

ⓐ $(0,0)$ ⓑ $(1,1)$ ⓒ $(-1,-1)$

→ Check the either local maximum or local minimum value of $f(x,y)$.

So that we have to find the value of D at ⓐ $(0,0)$.

$f_{xx} = ?$ $f_{yy} = ?$ and $f_{xy} = ?$

$f_x = 4x^3 - 4y$

$f_{xx} = 12x^2 - 0$ at $(0,0)$

$f_{xx}(0,0) = 12(0)^2 = 0$

$\boxed{f_{xx}(0,0) = 0}$

$f_y = 4y^3 - 4x$

$f_{yy} = 12y^2 - 0$ at $(0,0)$

$f_{yy}(0,0) = 12(0)^2 = 0$

$\boxed{f_{yy}(0,0) = 0}$

$f_x = 4x^3 - 4y$

$f_{xy} = 0 - 4y$ at $(0,0)$

$\boxed{f_{xy}(0,0) = -4}$

Now find $D = ?$ at $(0,0)$

$D(0,0) = f_{xx}(0,0)\ f_{yy}(0,0) - [f_{xy}(0,0)]^2$

$= (0)\ (0) - [-4]^2 = \boxed{-16}$

$\boxed{D(0,0) = -16} < 0$, $f_{xx} = 0$ (equal to zero)

So it follows ⓒ of 2nd derivative test,

So the points at origin $(0,0)$ is Saddle points.

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Now, find the value of D at $(1,1)$

$f_{xx} = ?$ $\quad f_{yy} = ?$ $\quad f_{xy} = ?$

$$f_x = 4x^3 - 4y$$
$$f_{xx} = 12x^2 - 0 \quad \text{at } (1,1)$$
$$f_{xx}(1,1) = 12(1)^2 = 12$$
$$\boxed{f_{xx}(1,1) = 12}$$

$$f_y = 4y^3 - 4x$$
$$f_{yy} = 12y^2 - 0 \quad \text{at } (1,1)$$
$$f_{yy}(1,1) = 12(1)^2 = 12$$
$$\boxed{f_{yy}(1,1) = 12}$$

$$f_x = 4x^3 - 4y$$
$$f_{xy} = 0 - 4y \quad \text{at } (1,1)$$
$$\boxed{f_{xy}(1,1) = -4}$$

find $D = ?$ at $(1,1)$

$$D(1,1) = f_{xx}(1,1)\; f_{yy}(1,1) - [f_{xy}(1,1)]^2$$
$$= (12)\;(12) - [-4]^2$$
$$= 144 - 16$$
$$\boxed{D(1,1) = 128} > 0 \qquad f_{xx} = 12 > 0$$

So it follows (a) of 2nd derivative test,

~~[illegible] is local minimum.~~

So the local mini value of $f(x,y)$ should

$$f(1,1) = x^4 + y^4 - 4xy + 1 \quad \text{at } (1,1)$$
$$= (1)^4 + (1)^4 - 4(1)(1) + 1$$
$$= 1 + 1 - 4 + 1$$
$$= 3 - 4$$
$$\boxed{f(1,1) = -1} \rightarrow$$

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find the value of D at (-1 , -1)

$f_{xx} = ?$ $f_{yy} = ?$ $f_{xy} = ?$

$f_x = 4x^3 - 4y$

$f_{xx} = 12x^2 - 0$ at $(-1, -1)$

$f_{xx}(-1,-1) = 12(-1)^2 = 12$

$\boxed{f_{xx}(-1,-1) = 12}$

$f_y = 4y^3 - 4x$

$f_{yy} = 12y^2 - 0$ at $(-1, -1)$

$f_{yy} = (-1,-1) = 12(-1)^2 = 12$

$\boxed{f_{yy} = (-1,-1) = 12}$

$f_{x} = 4x^3 - 4y$

$f_{xy} = 0 - 4y$ at $(-1, -1)$

$\boxed{f_{xy} = (-1,-1) = -4}$

find $D = ?$ at $(-1, -1)$

$D(-1,-1) = f_{xx}(-1,-1) \; f_{yy}(-1,-1) - [f_{xy}(-1,-1)]^2$

$= (12) \quad (12) - [-4]^2$

$= 144 - 16$

$\boxed{D(-1,-1) = 128} > 0$ $f_{xx} = 12 > 0$

It follows (a) of 2nd derivative test.

So the local minimum value of $f(x,y)$ should be

$f(-1,-1) = x^4 + y^4 - 4xy + 1$ at $(-1, -1)$

$= (-1)^4 + (-1)^4 - 4(-1)(-1) + 1$

$= 1 + 1 - 4 + 1$

$\boxed{f(-1,-1) = -1}$ → is also a local minimum.